V2V-Voice2Visuals: Instant Voice-to-Sign Language Translation for Seamless Accessibility and Communication

ABSTRACT

V2V – Voice2Visuals presents a machine learning–based mobile application that enables real-time translation of spoken voice into sign language visuals to enhance communication accessibility for individuals with hearing impairments. The proposed system captures voice input through a mobile device, converts it into text using speech recognition techniques, and processes the text using natural language processing models. The processed output is then mapped to corresponding sign language representations and displayed as visual animations on the mobile interface. The system integrates machine learning algorithms with mobile application development to achieve accurate, low-latency, and user-friendly performance. By providing instant voice-to-sign language translation, the application minimizes communication barriers in everyday scenarios such as education, healthcare, and public services. Experimental evaluation demonstrates that the proposed approach offers effective real-time translation while maintaining usability and accessibility. V2V – Voice2Visuals contributes toward inclusive communication by leveraging intelligent systems and mobile technologies.

Submitted by:

M. Tejaswini(23UP1A05G0)

M. Lahari(23UP1A05F4)

G. Yashaswini(23UP1A05E3)

S.Archana(23UP1A05H9)